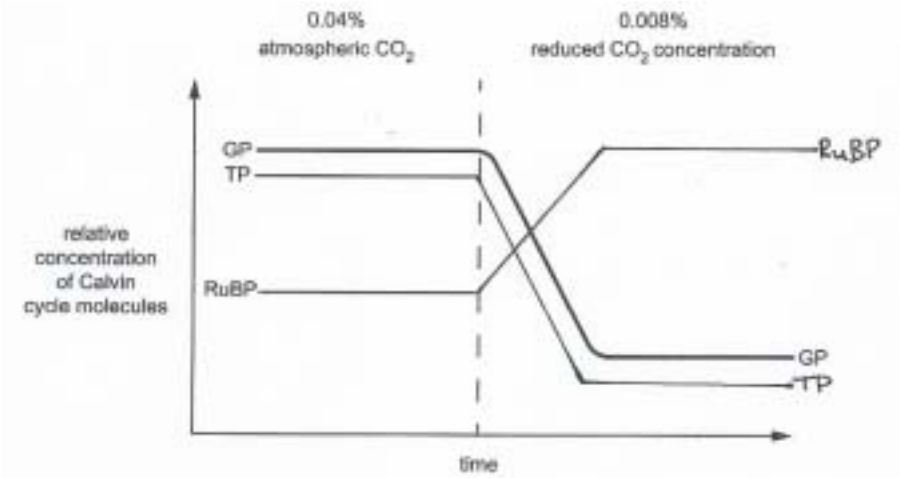


2(a)(i)	any two from: 1 ref. to light energy / photons ; 2 for, light-dependent stage / photophosphorylation / photolysis / photoexcitation / photoactivation ; 3 to make, reduced NADP / ATP ; 4 to open stomata (for CO₂ to enter) ;	2
2(a)(ii)	correctly showing that TP reduces to a plateau below GP ; correctly showing that RuBP increases to a plateau ;  The graph illustrates the response of Calvin cycle molecules to a sudden reduction in CO₂ concentration. When CO₂ is reduced from 0.04% to 0.008%, the concentration of GP and TP decreases significantly, while the concentration of RuBP increases. This occurs because RuBP is used to fix CO₂ and GP/TP are the products of this reaction. With less CO₂, less RuBP is consumed and more GP/TP is produced, leading to the observed changes in their relative concentrations.	2

Question	Answer	Marks
2(a)(iii)	1. decrease in / no, CO₂, diffusing / moving, to, air spaces / palisade cells / mesophyll cells / chloroplasts / stroma ; <i>any two from:</i> 2 stress response / AW ; 3 <i>ref. to</i> abscisic acid / ABA ; 4 <i>ref. to</i> action of Ca²⁺ as a second messenger ; 5 stomata close ;	3
2(b)(i)	$\frac{28 - 20.5}{20.5} \times 100$ or $\frac{7.5}{20.5} \times 100$; A 20.4 or 20.6 A 37.3 or 35.9 <i>answer to 1 decimal place</i> 36.6 (%) ; <i>apply ecf</i>	2

Question	Answer	Marks
2(b)(ii)	<i>any four from:</i> <i>no mention of increased or more – ecf and max 3</i> 1 Sox4 plants have an extra copy of <i>SBPase</i> gene ; 2 (overall) increased, expression / transcription, of <i>SBPase</i> gene (so more <i>SBPase</i>) ; 3 increased (rate of) / more, regeneration / production, of RuBP ; 4 (so) increased / more, carbon fixation / Calvin cycles / light independent reaction / TP / GP ; 5 more, glucose / sucrose, for respiration / ATP production or more, starch / lipid, for storage or more cellulose for cell walls ; 6 more, amino acids / proteins, for growth ; 7 more, cell division / cell elongation / mitosis ; 8 AVP ; e.g.(new) <i>SBPase</i> / <i>SBPase</i> gene, more efficient	4
2(b)(iii)	<i>nitrogen needed to produce:</i> nucleotides for, transcription (of <i>SBPase</i> gene) / expression (of <i>SBPase</i> gene) / mRNA production ; amino acids for, protein synthesis / <i>SBPase</i> production / enzyme production ;	2